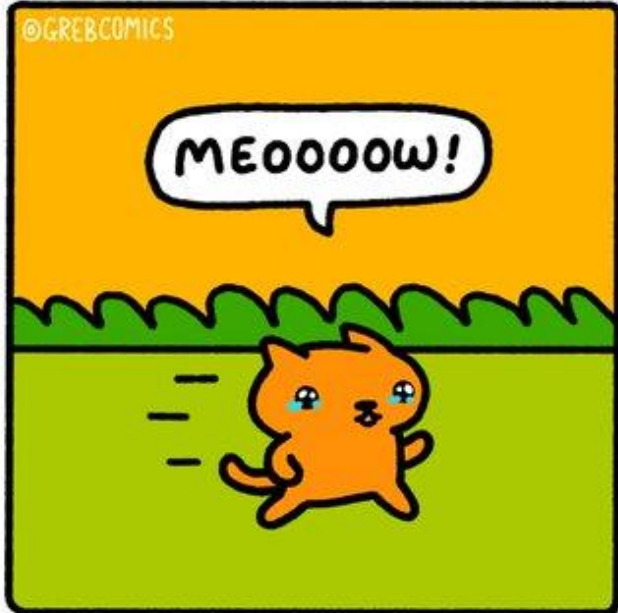




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Taxes and subsidies

Chapter 6

Objectives

- To understand taxes and subsidies
 - What they mean
 - How they affect supply and demand
 - How they affect welfare

Commodity tax: introduction

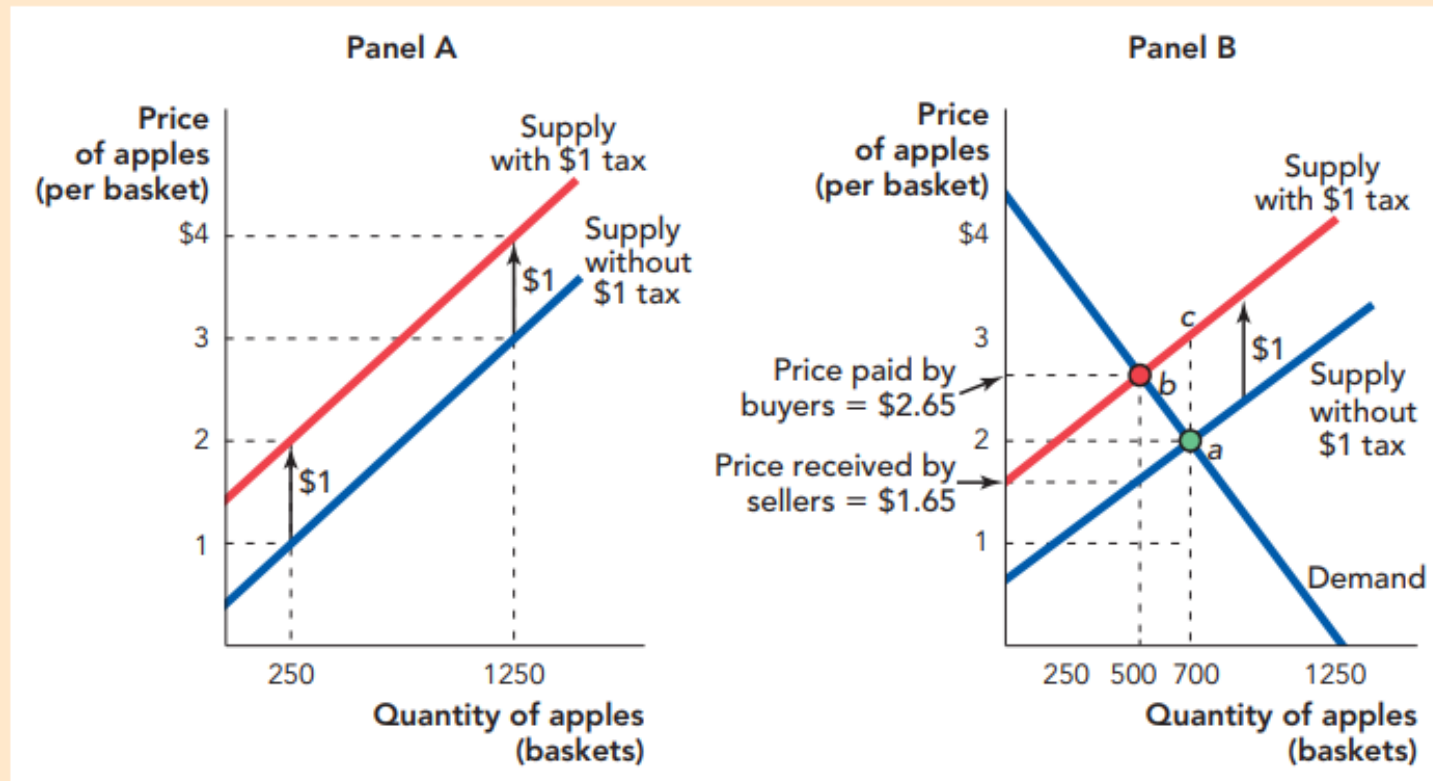
- A commodity tax creates a wedge between what buyers pay (P_b) and what sellers receive (P_s) and the difference is equal to the tax (T) .
- In terms of economic outcomes, **it does not matter who pays: the sellers or the buyers**

Commodity tax: suppliers and buyers

- A (commodity) tax can be levied on the **suppliers**. In this case, it shifts the supply curve “**up**” by the amount of the tax.
 - Why? The tax increases the **total** marginal cost of production -- the firm has to pay its marginal production cost and its tax “liability”.
- A (commodity) tax can also be levied on the **buyers**. In this case, it shifts the demand curve “**down**” by the amount of the tax.
 - Why? The demand curve represents the buyer’s willingness to pay. If a tax is imposed, the buyer is **willing to pay the producer less** because she has a **tax “liability”** she must pay, so that the **total amount** that buyers are willing to pay **including the tax** is the same.

Commodity tax: suppliers

FIGURE 6.1



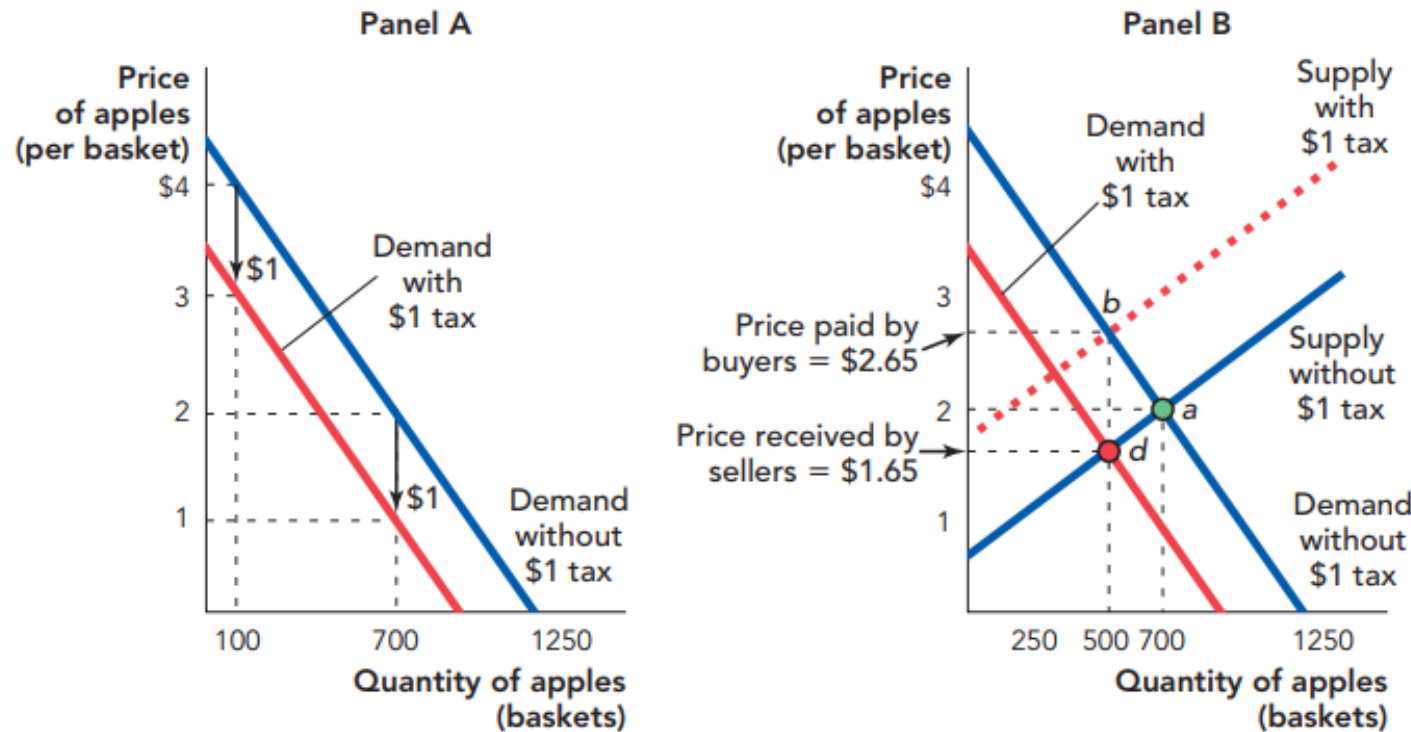
A Tax on Apple Sellers

Panel A: A \$1 tax on apple sellers shifts the supply curve up by \$1.

Panel B: A \$1 tax on apple sellers shifts the supply curve up by \$1, changing the equilibrium from point *a* to point *b*.

Commodity tax: buyers

FIGURE 6.2



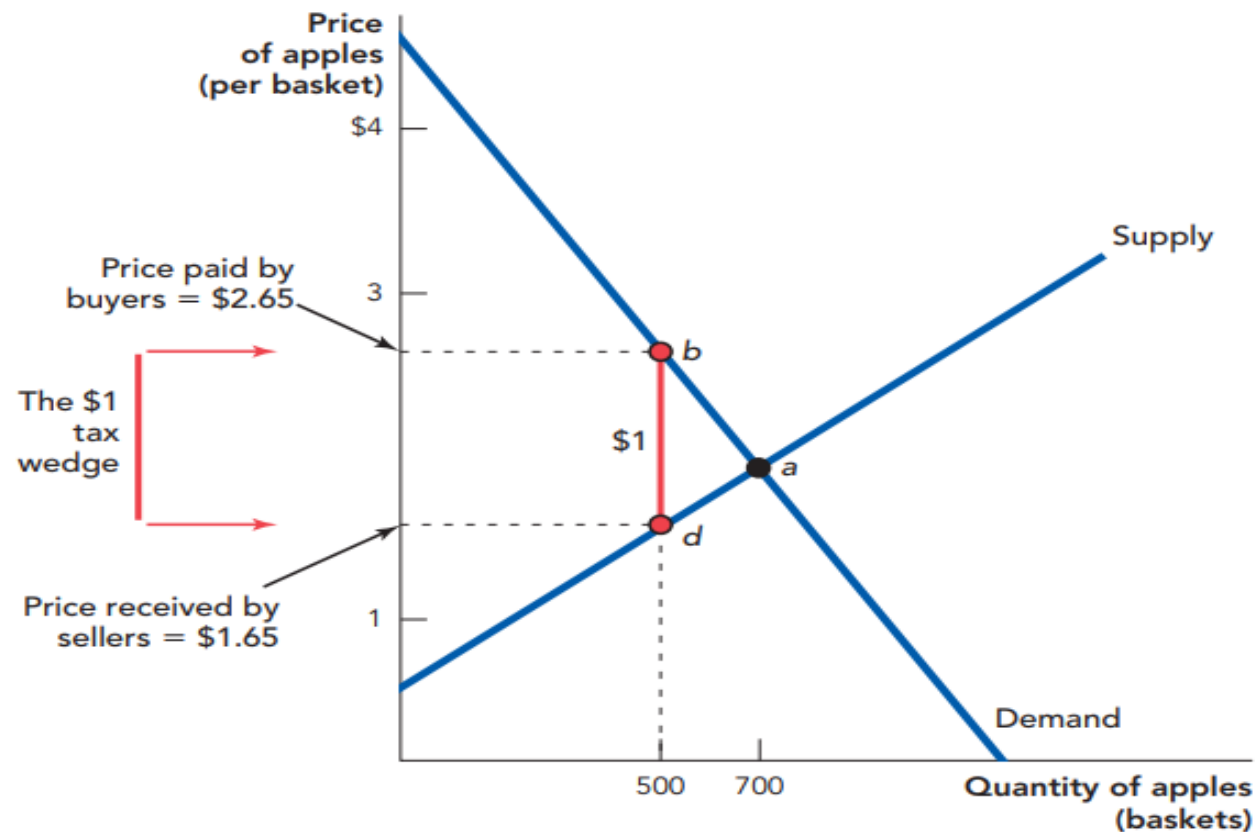
A Tax on Apple Buyers

Panel A: A \$1 tax on apple buyers shifts the demand curve down by \$1.

Panel B: A \$1 tax on apple buyers shifts the demand curve down by \$1, changing the equilibrium from point a to point d.

The Tax Wedge

FIGURE 6.3



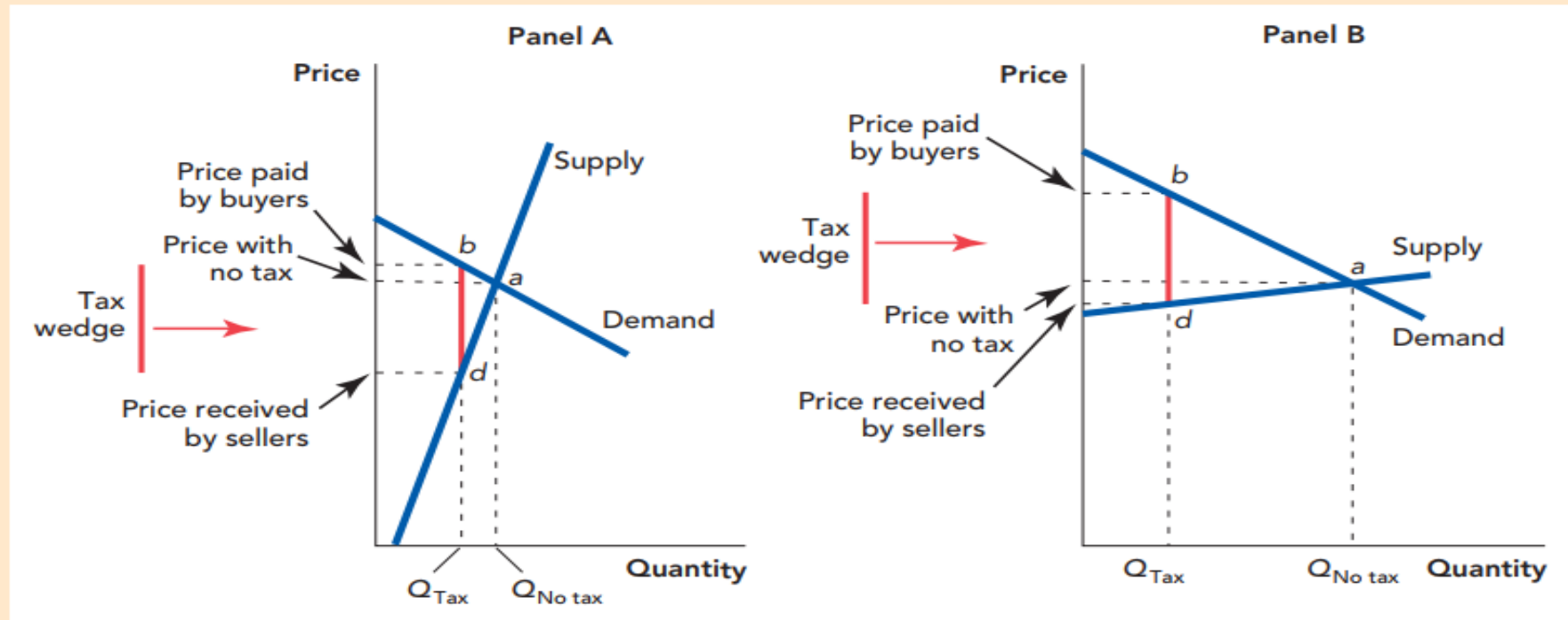
The Tax Wedge If the tax is \$1, the price paid by the buyers must be \$1 higher than the price received by the sellers. Driving a \$1 tax wedge into the diagram shows us the new equilibrium must be where the price paid by the buyers is \$2.65, the price received by the sellers is \$1.65, and the quantity traded is 500.

The Tax Wedge

- We now want to measure the **tax burden** or the **tax incidence**.
- The **tax incidence**, or **tax burden** can be considered by going back to the original equilibrium price.
- The tax burden, in terms of price, that falls on consumers is $b - a$
- The tax burden, in terms of price, that falls on sellers is $a - d$
- Total burden, in terms of price, is $b - d = T$

Elastic vs Inelastic Supply and Demand

FIGURE 6.4



The More Elastic Side of the Market Can Escape More of the Tax

Panel A: When demand is more elastic than supply, buyers pay less of the tax than sellers.

Panel B: When supply is more elastic than demand, suppliers pay less of the tax than buyers.

Elastic vs Inelastic Supply and Demand

- When demand is more elastic than the supply, buyers pay less of the tax than sellers.
 - Why? Think about cereal brands, generic drugs.
 - Which kind of consumers are more likely to switch?
 - Can you think of more examples?
- When supply is more elastic than the demand, sellers pay less of the tax than buyers.
 - Why? Think about prescription drugs, baby formula.
 - Which kind of suppliers are more likely to switch?
 - Can you think of more examples?

Elastic vs Inelastic Supply and Demand

- When the demand curves and/or the supply curves are relatively more **inelastic**, the government is more likely to raise more government revenues from taxes.

Taxes on cigarettes

- Taxes on cigarettes vary widely across states.
- For example, in Alabama and in South Carolina taxes on cigarettes are \$0.86 and \$0.88, respectively. Whereas in New York and Massachusetts, cigarettes are taxed at \$4.75 and \$4.10.
- The demand for cigarettes is likely to be inelastic. Why?
- The supply for cigarettes in any state is likely to be much more elastic. Why?
- Who is likely to pay the burden of the tax?

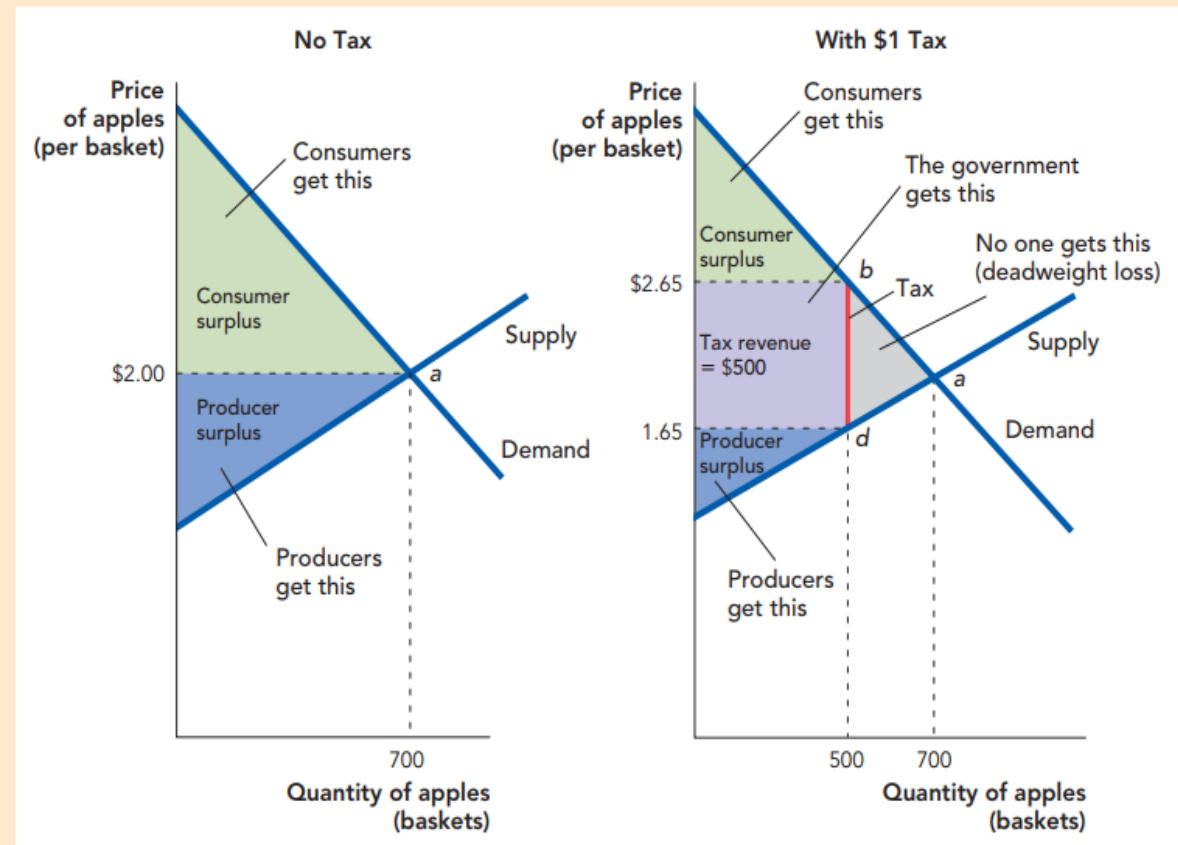
Deadweight loss

Given that costs (burden) of taxes can fall on **buyers** and **sellers**, it would be nice if we had a way to **calculate the costs**. Luckily, we do.

Recall **consumer** and **producer surplus**.

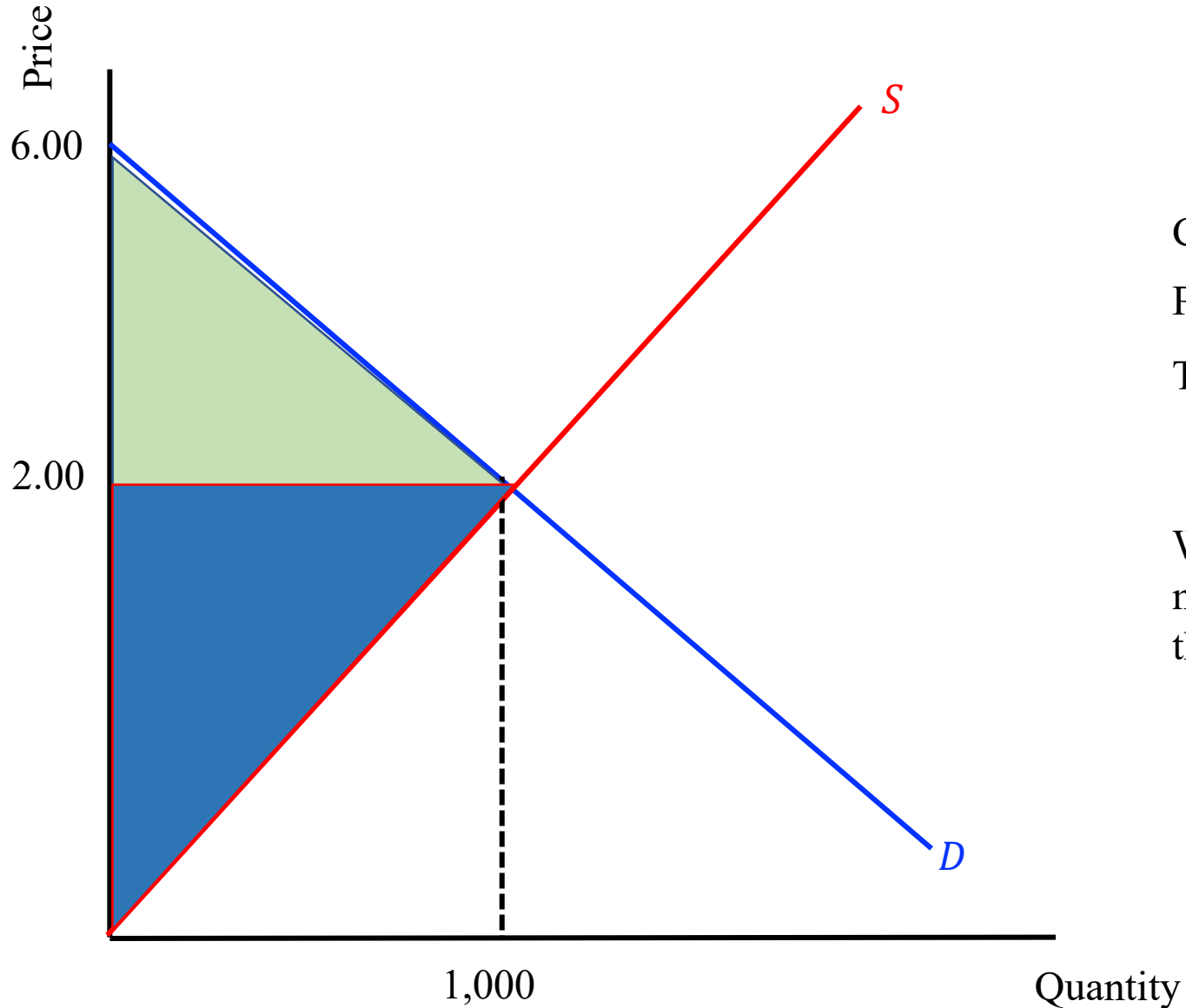
When taxes are imposed, consumer and producer surplus shrinks!

FIGURE 6.5



A Tax Generates Revenue and Creates a Deadweight Loss With no tax, producer plus consumer surplus is maximized in the left panel. With the tax, consumer surplus and producer surplus are smaller and tax revenues are larger. But tax revenues increase by less than producer and consumer surplus fall. As a result, the tax creates the deadweight loss shown by the gray area (triangle *abd*) in the right panel.

Consumer and Producer Surplus



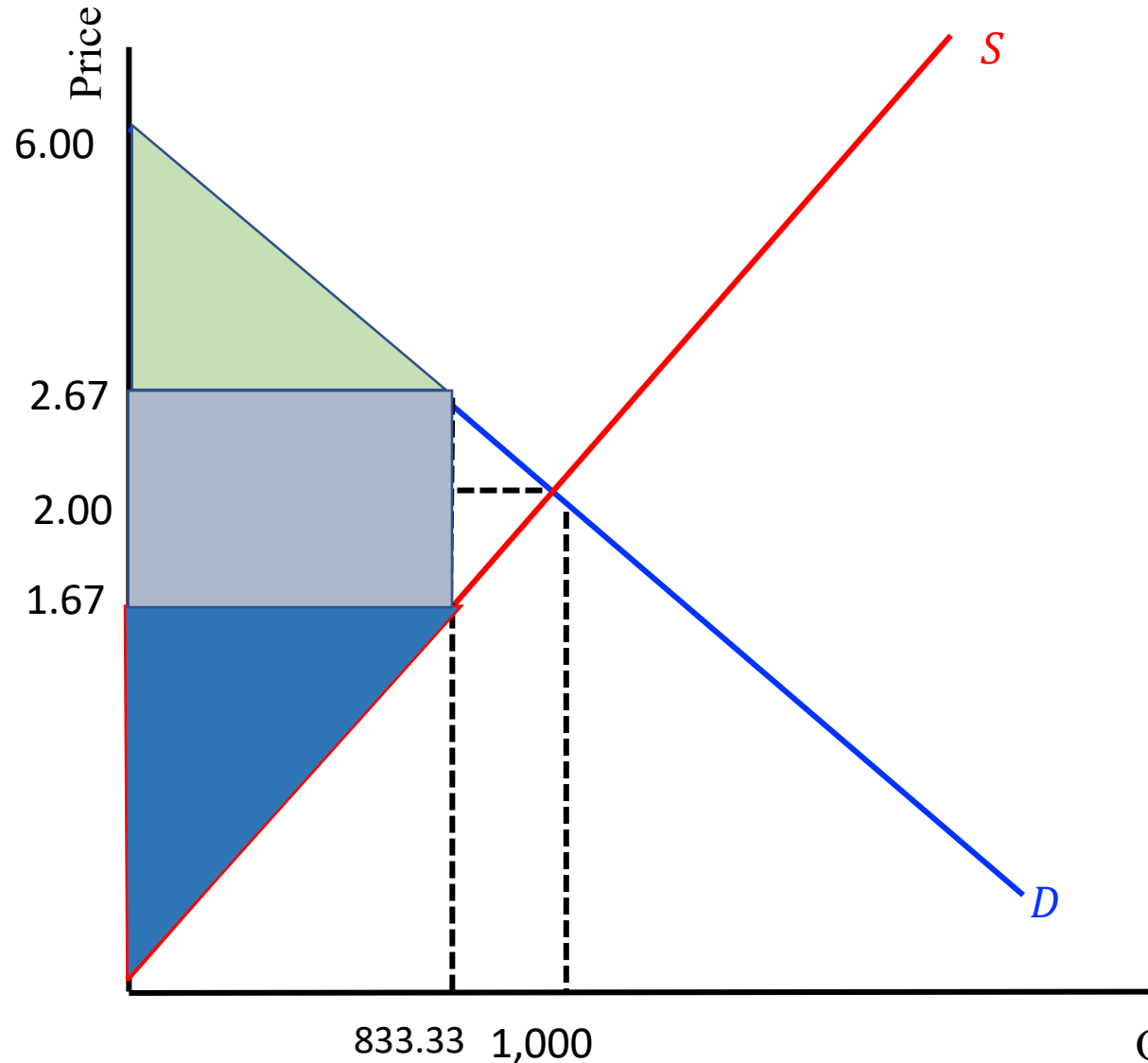
$$\text{Consumer Surplus} = \frac{1}{2}(6 - 2)1,000 = 2,000$$

$$\text{Producer Surplus} = \frac{1}{2}(2 - 0)1,000 = 1,000$$

$$\text{Total Surplus} = \$3,000$$

With competitive markets, **total surplus** was maximized at the point of equilibrium -- and that this was indicative of *efficiency*

Welfare Implications of tax



In the new equilibrium, the buyers pay a total of **\$2.67** per gallon of gasoline (**\$2.67 = \$1.67 + \$1.00**).

Consumer Surplus = $\frac{1}{2}(6.00 - 2.67)833.33 = \$1,388.88$

In the new equilibrium, the seller receives **\$1.67** per gallon of gasoline.

Producer Surplus = $\frac{1}{2}(1.67 - 0.00)833.33 = \694.44

Government Gain in Revenues = \$833.33

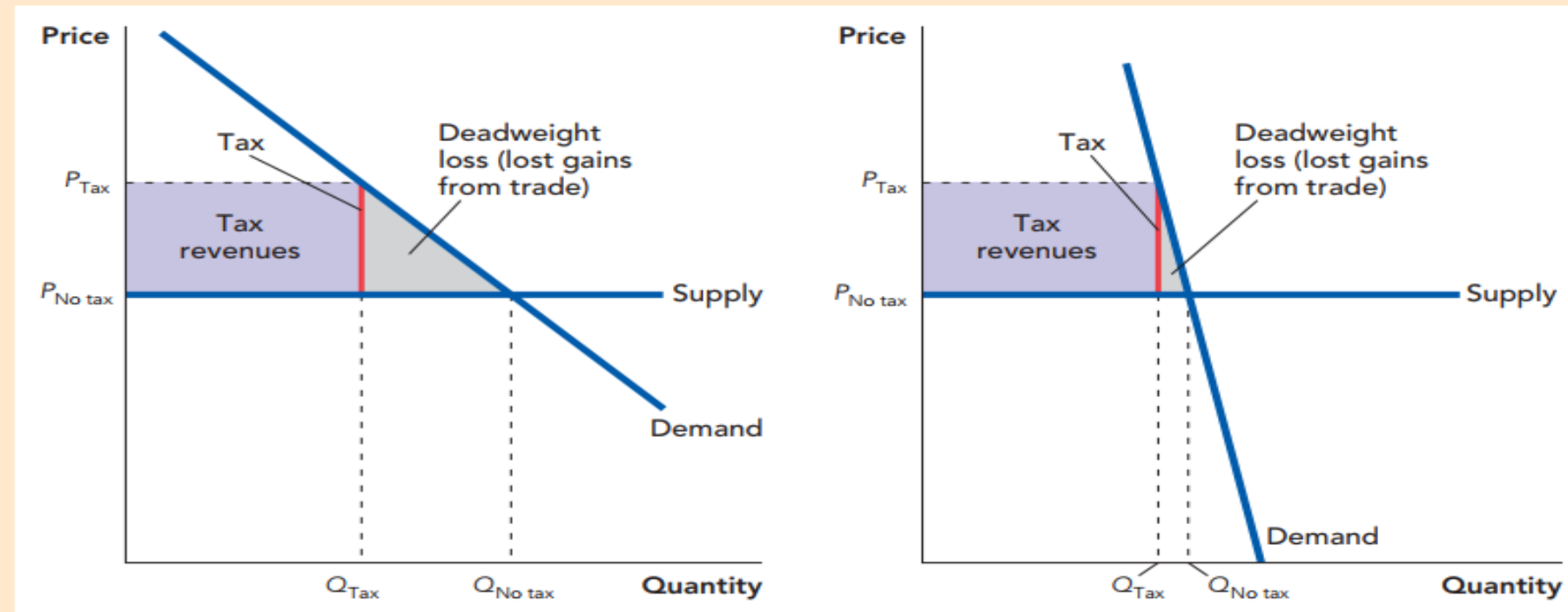
With taxes, we can still measure what happens to **consumer** and **producer** surplus after the **tax**. Recall with taxes there is a third actor – the **government** and the **government** receives **revenue** from the **taxes**.

Welfare Implications of tax

- Not all of the loss of consumer surplus was “transferred” to the government.
- Not all of the loss of producer surplus was “transferred” to the government.
- The area of producer surplus and consumer surplus that is “completely” lost after the tax is referred to as the **deadweight loss**.
- Deadweight loss is a measure of how **far the total surplus** departs from the **efficient** outcome

Deadweight Loss: Elastic vs Inelastic Demand

FIGURE 6.6



The Deadweight Loss from Taxation Is Larger the More Elastic the Demand Curve The tax rate and tax revenues are the same but the deadweight loss is larger in the left panel where the demand curve is more elastic.

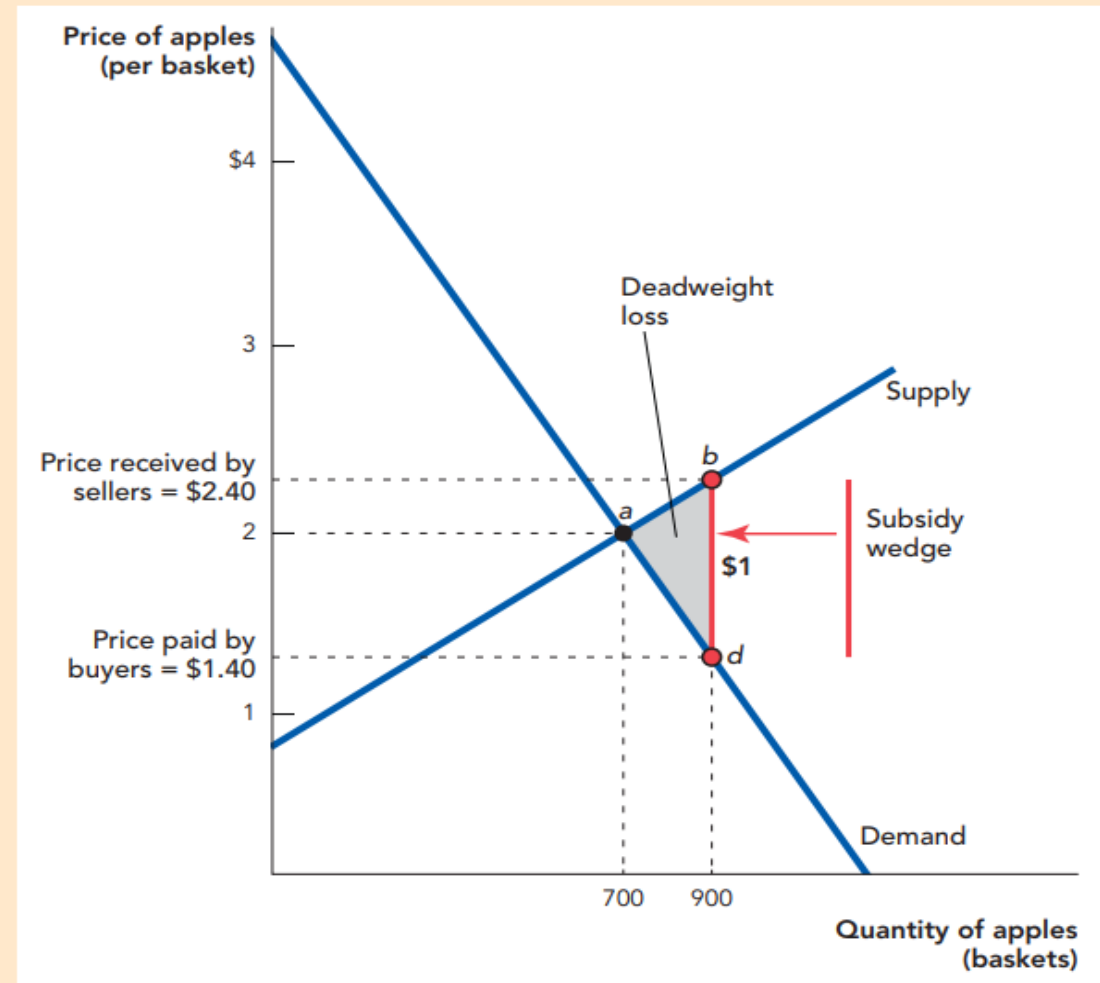
Deadweight Loss: Elastic vs Inelastic Demand

- Deadweight losses arise when there are **distortions** (as in the case of a tax) and when there are **inefficiencies**.
 - There is a **distortion** with a tax because a tax drives a **wedge** between what buyers pay and what sellers receive.
- Both buyers and sellers will bear some of the burden from **efficiency loss** from a taxation.
 - If demand is relatively inelastic, the **deadweight loss** is **lower** than if the demand curve is relatively elastic.
 - If supply is relatively inelastic, the **deadweight loss** is lower than if the supply curve is relatively elastic.

Subsidy wedge

A **subsidy** drives a wedge between the price received by the sellers and the price paid by the buyers. A subsidy also creates a **deadweight loss**.

FIGURE 6.7



The Subsidy Wedge A subsidy drives a wedge between the price received by the sellers and the price paid by the buyers. A subsidy creates a deadweight loss (triangle *abd*).

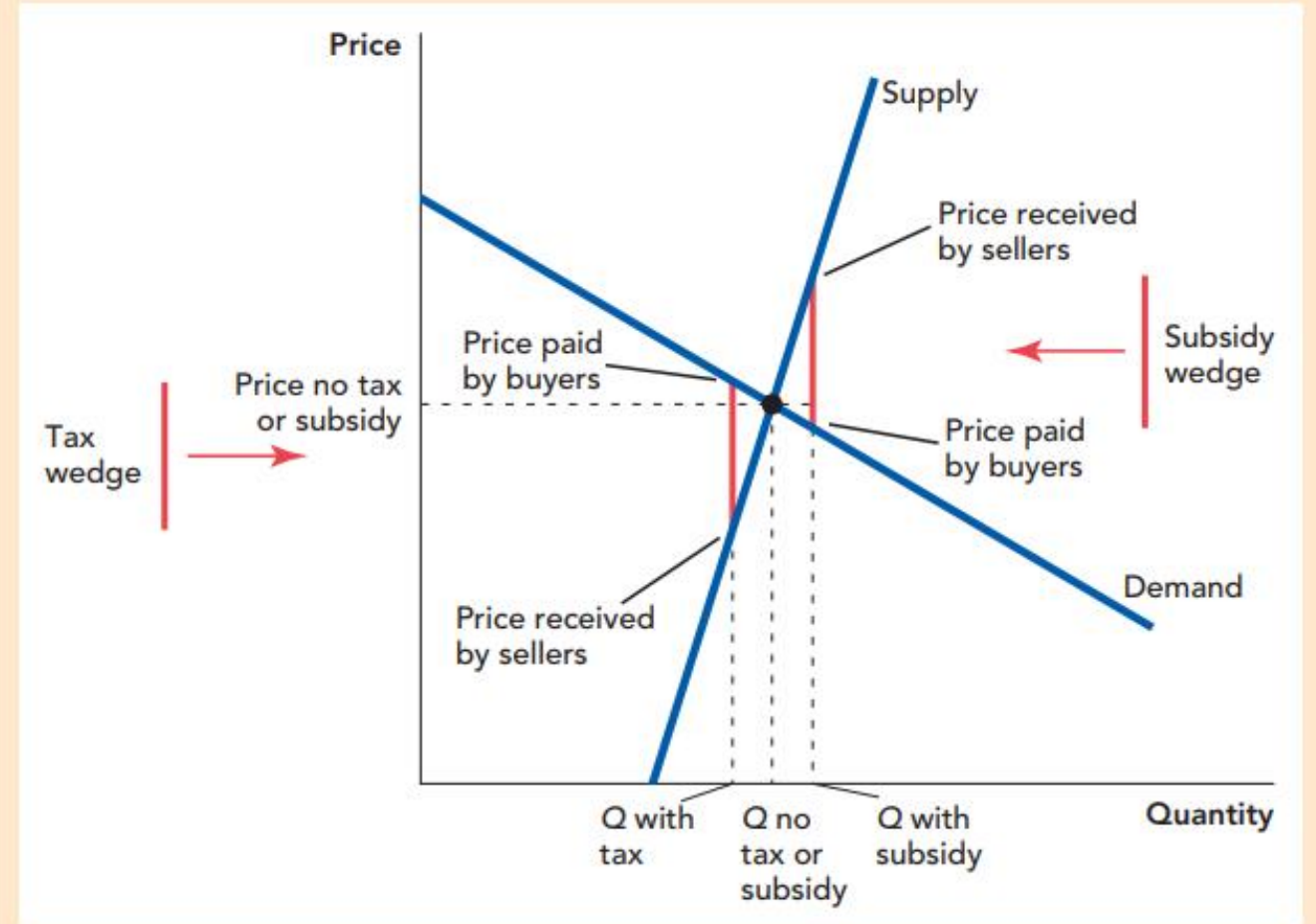
Subsidies

- Who gets the subsidy does not depend on who gets the check from the government.
- Who benefits from a subsidy does depend on the elasticities of demand and supply.
 - If demand is relatively more inelastic, buyers receive more of the benefits from the subsidy.
 - If supply is relatively more inelastic, sellers receive more of the benefits from the subsidy.
- Subsidies must be paid for by taxpayers and they create an inefficient increase in transactions (trade).

Tax or Subsidy Burden

- When demand is more elastic than supply, suppliers bear more of the burden of a tax and receive more of the benefit of the subsidy.
- The more elastic side of the market will receive a smaller share of the subsidy (smaller benefit).
- The less elastic (more inelastic) side of the market will receive a greater share of the subsidy (greater benefit)

FIGURE 6.8



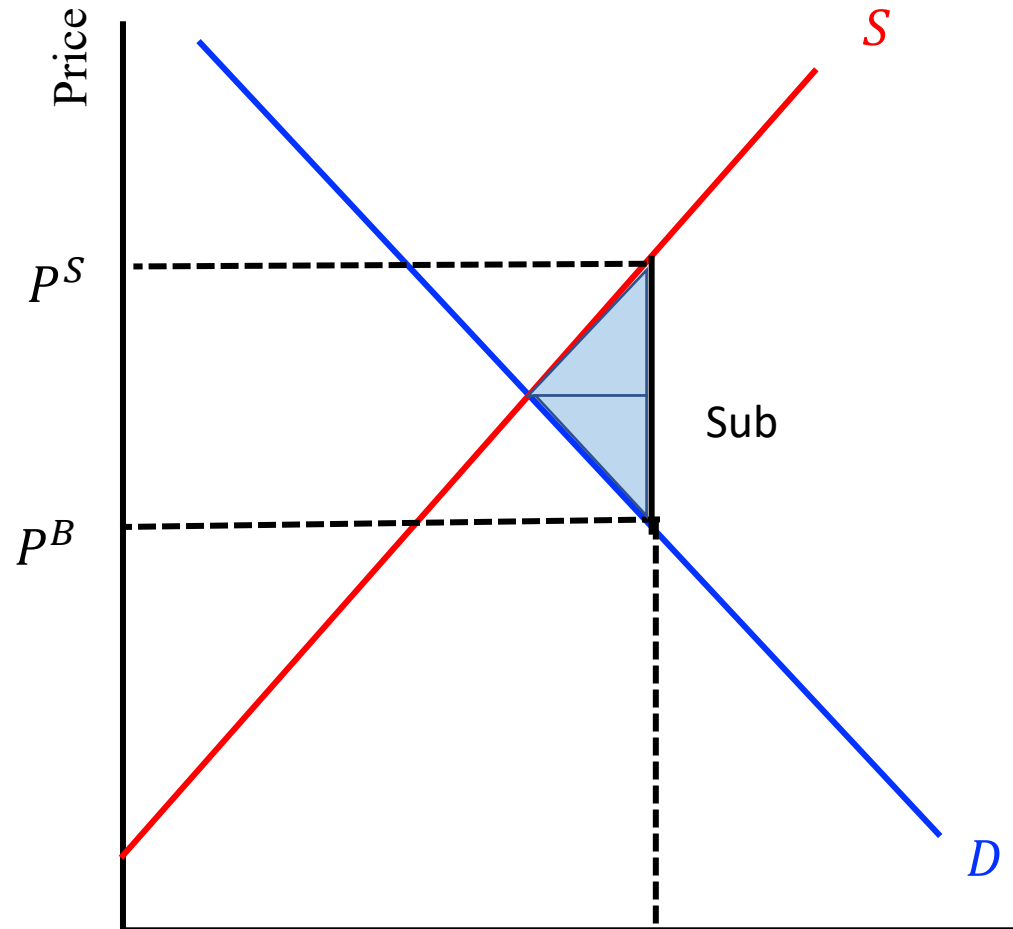
Whoever Bears the Burden of a Tax Receives the Benefits of a Subsidy

When demand is more elastic than supply, suppliers bear more of the burden of a tax and receive more of the benefit of a subsidy.

Subsidy Example: Rental Housing

- Now let's assume that we offer subsidies to the renter or to families that are looking to rent an apartment. How will this impact the market for apartment rentals?

Subsidy Example: Rental Housing



The subsidy increases the number of housing units for rent.

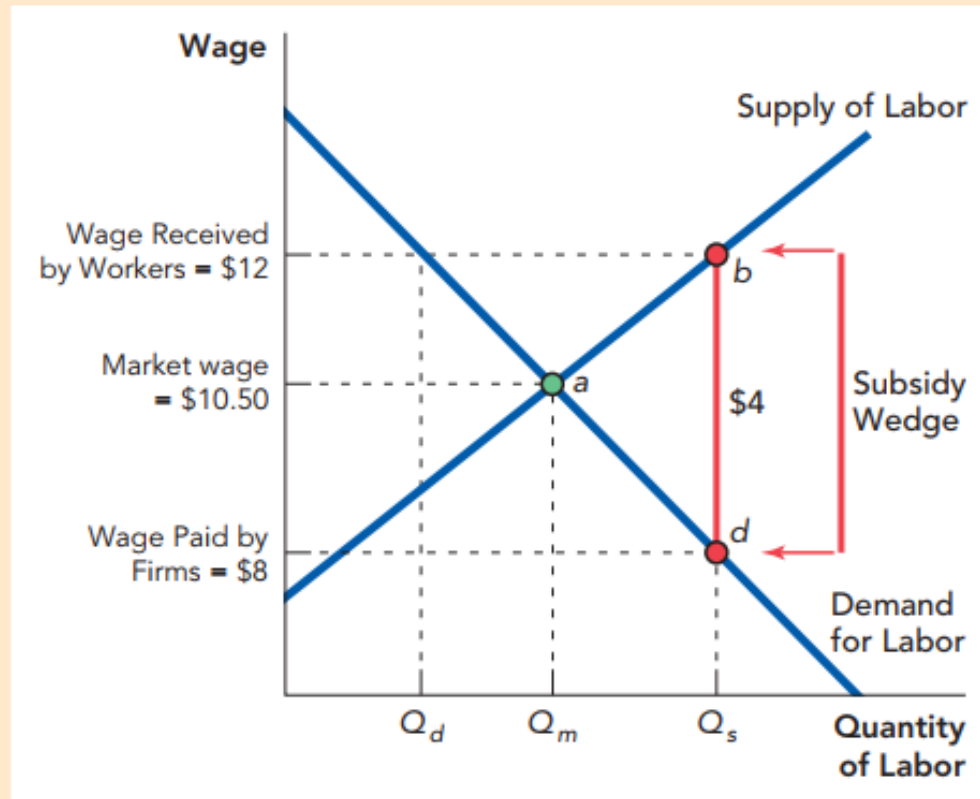
Consumers pay a **lower amount** out of pocket and renters receive a **higher amount**

The cost of the program is the subsidies that **must be paid** by the government

Q^S Quantity of Apartments

Earned Income Subsidy Schemes

FIGURE 6.9



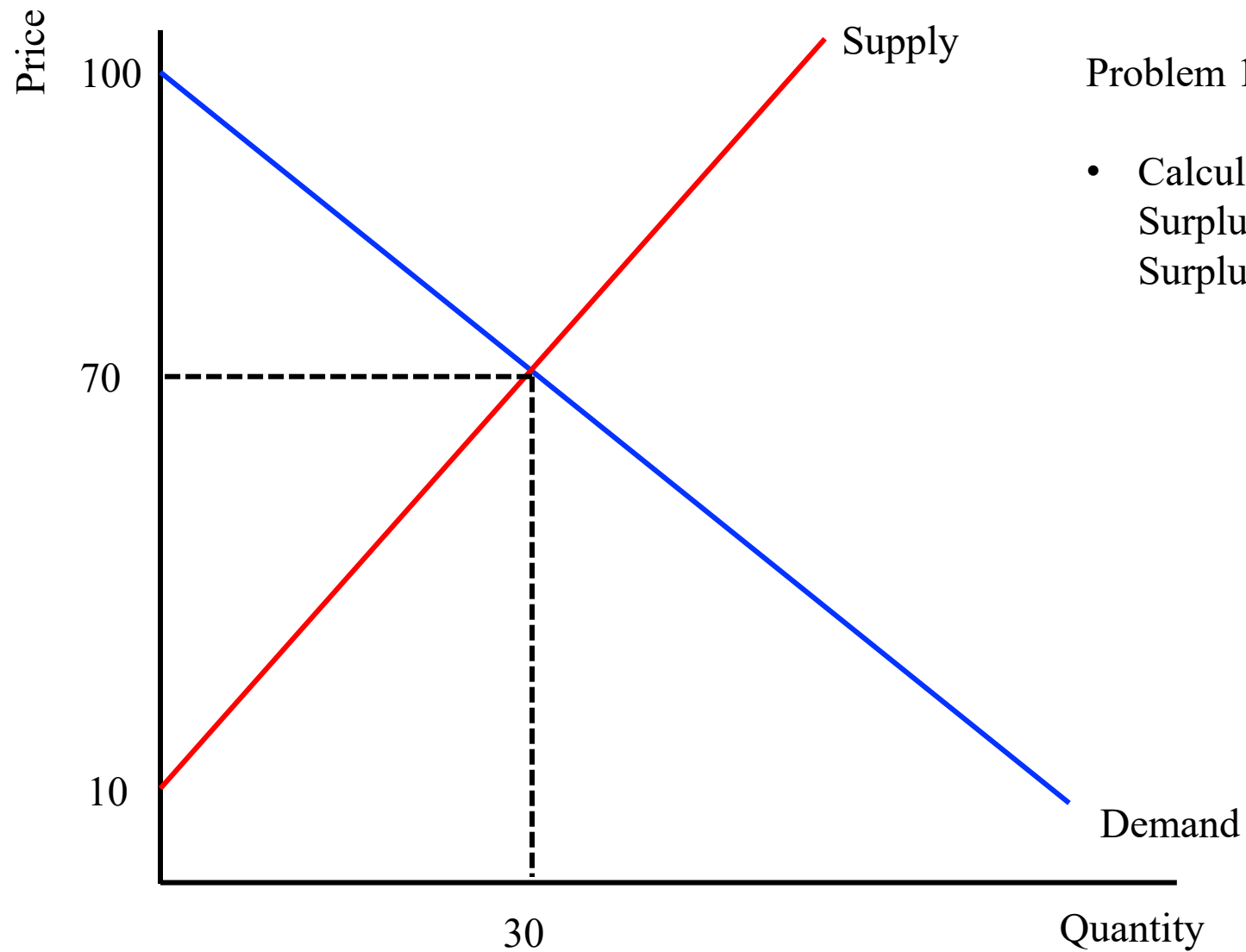
A Subsidy to Wages Increases Employment

The free market wage for low-skill workers is \$10.50. A subsidy to firms that hire low-wage workers increases the wage received by workers to \$12 and reduces the wage paid by firms to \$8. As a result, employment increases from Q_m to Q_s . The cost of the subsidy to the government is the subsidy amount, \$4, times the number of workers employed, Q_s .

Minimum wage subsidy schemes

- **Wage subsidy schemes** offer subsidies to employers who take on unemployed individuals from target groups for defined periods.
 - Target groups are often described by age and duration of unemployment, but other characteristics can be included.
- The reduction in unemployment is not equivalent, however, to the number of subsidized job placements.
- Both **deadweight losses** and **displacement** effects must be considered when assessing the impact.
 - Displacement occurs where workers from the target group are recruited in preference to others, thus altering the composition of unemployment, rather than reducing it.

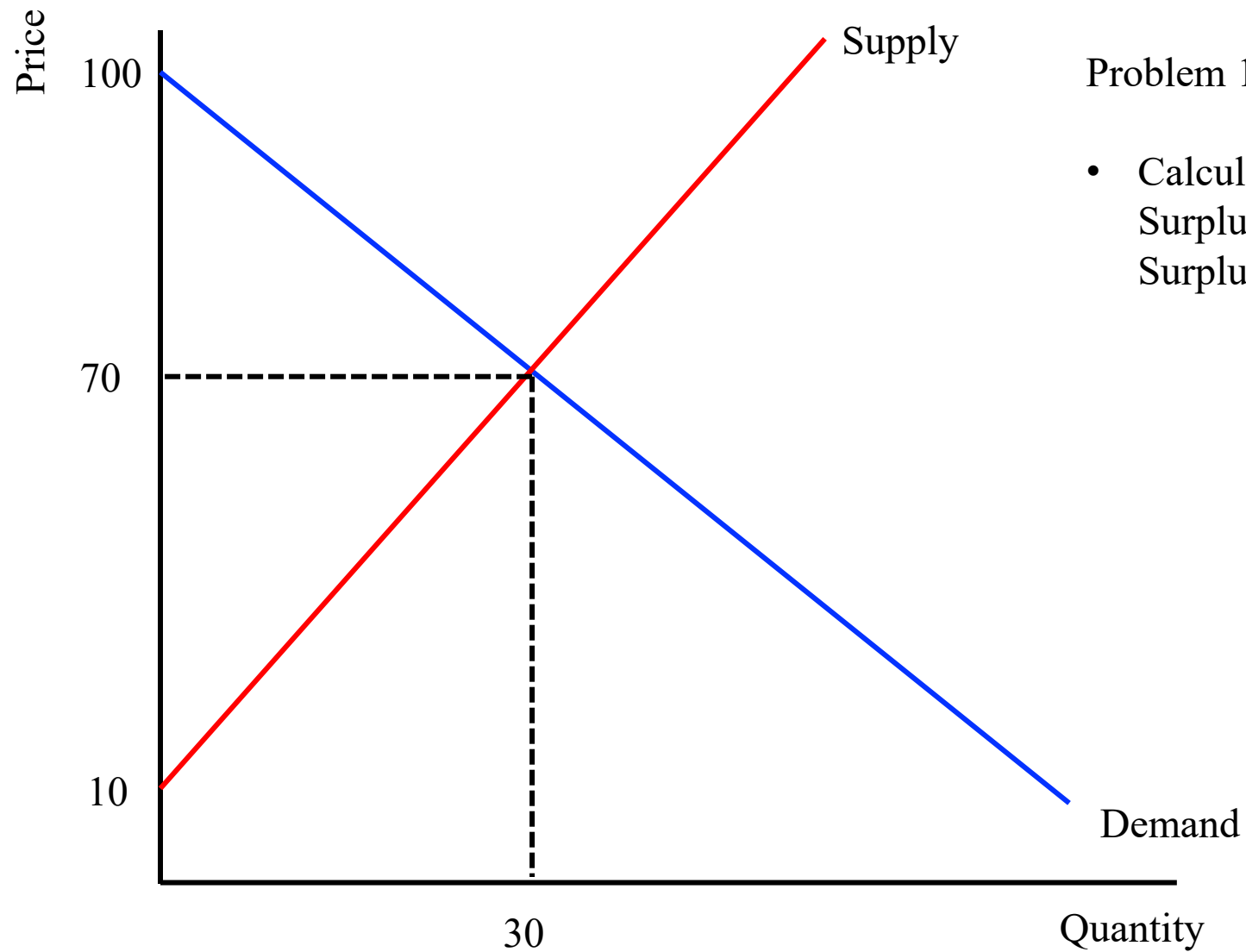
Practice Problems



Problem 1

- Calculate Producer Surplus and Consumer Surplus

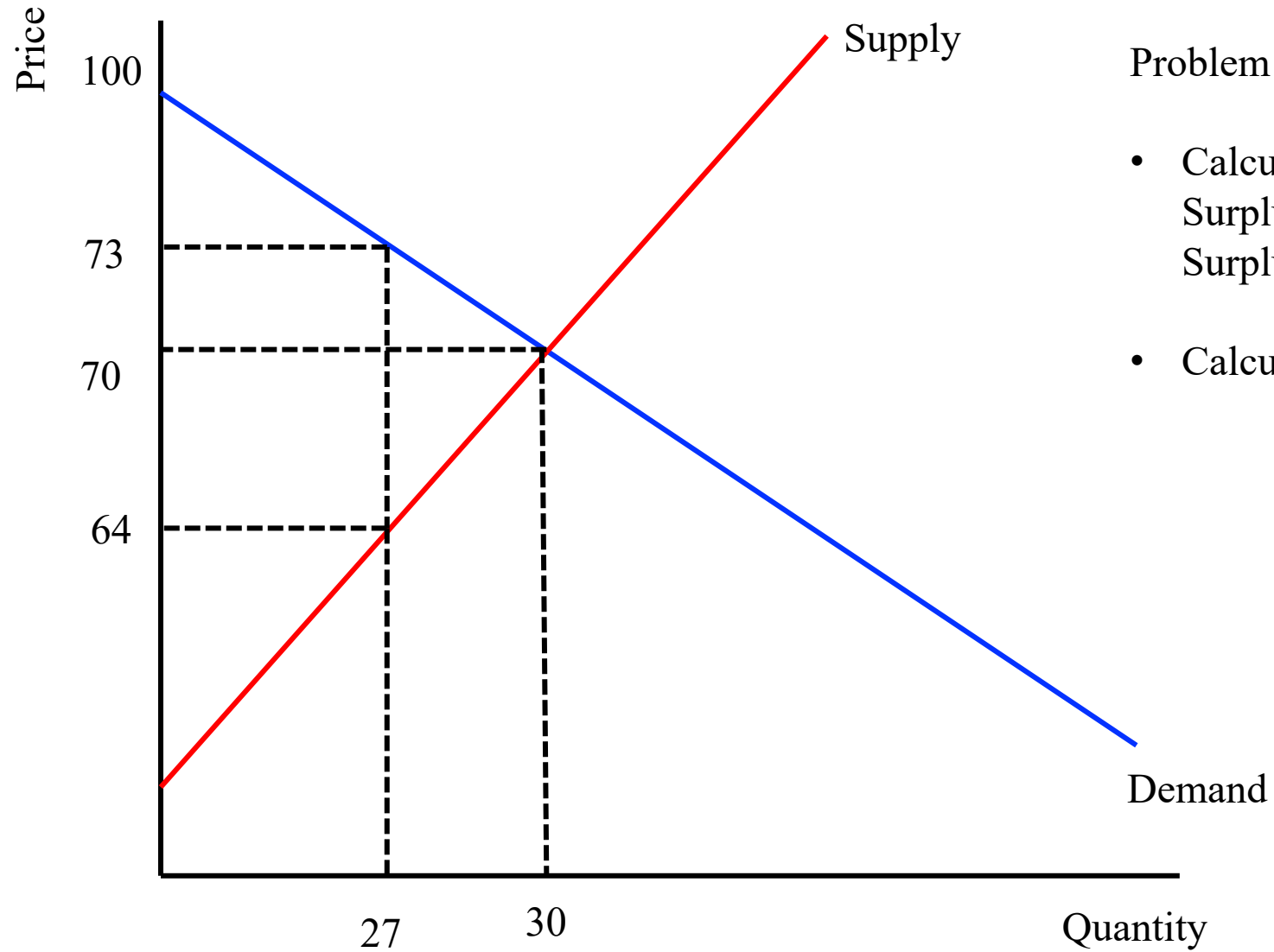
Practice Problems



Problem 1

- Calculate Producer Surplus and Consumer Surplus

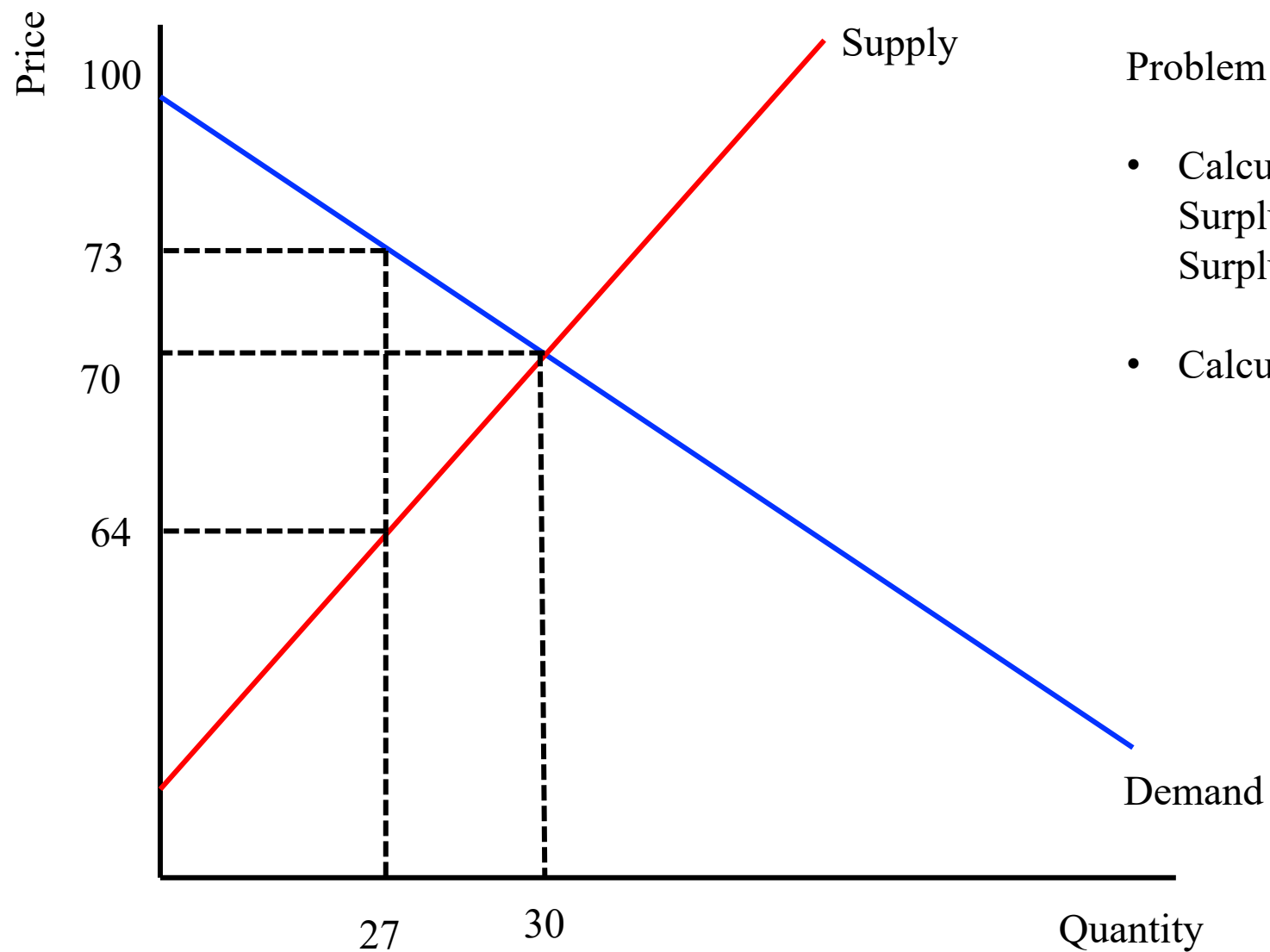
Practice Problems



Problem 1

- Calculate Producer Surplus and Consumer Surplus
- Calculate Tax Revenue

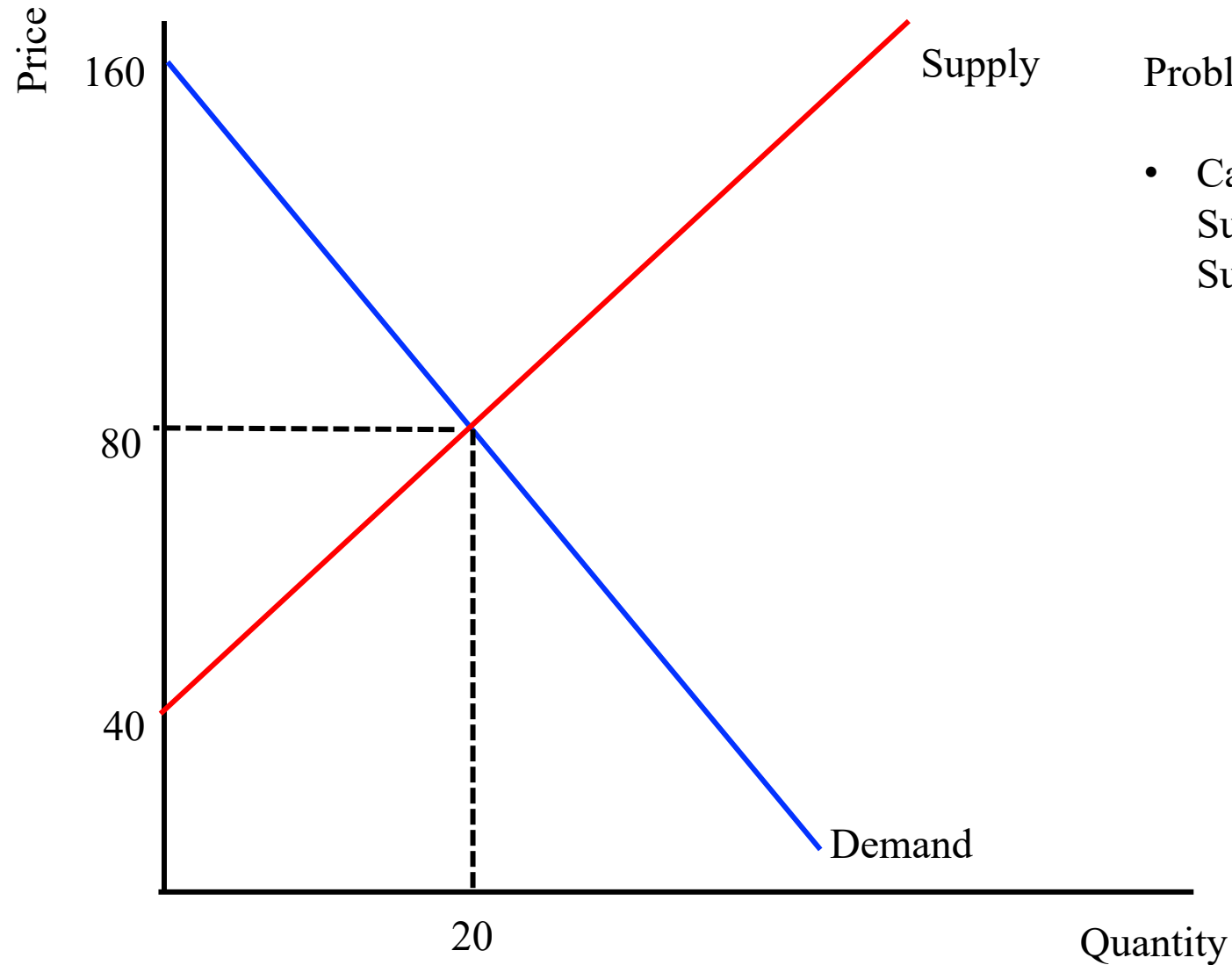
Practice Problems



Problem 1

- Calculate Producer Surplus and Consumer Surplus
- Calculate Tax Revenue

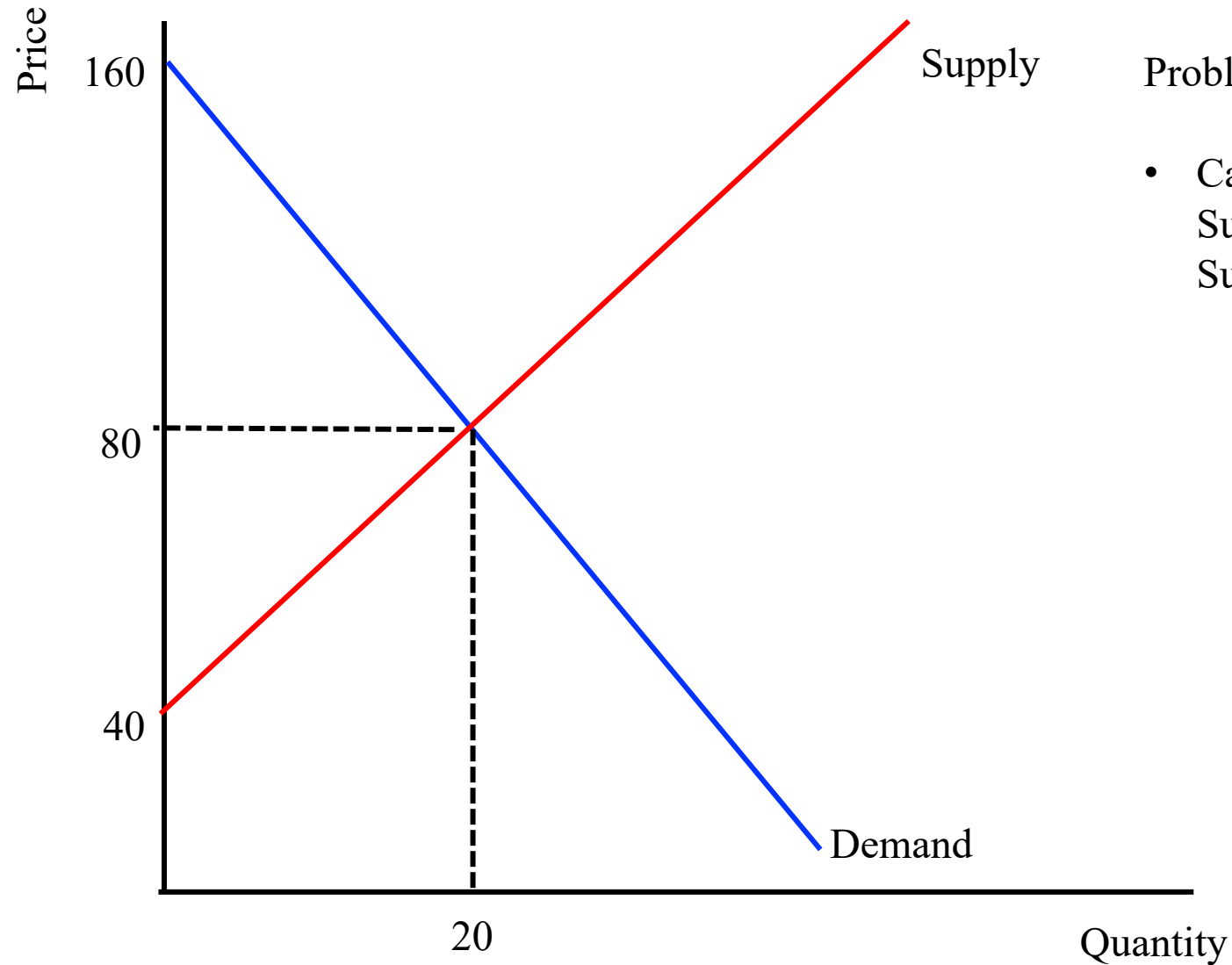
Practice Problems



Problem 2

- Calculate Producer Surplus and Consumer Surplus

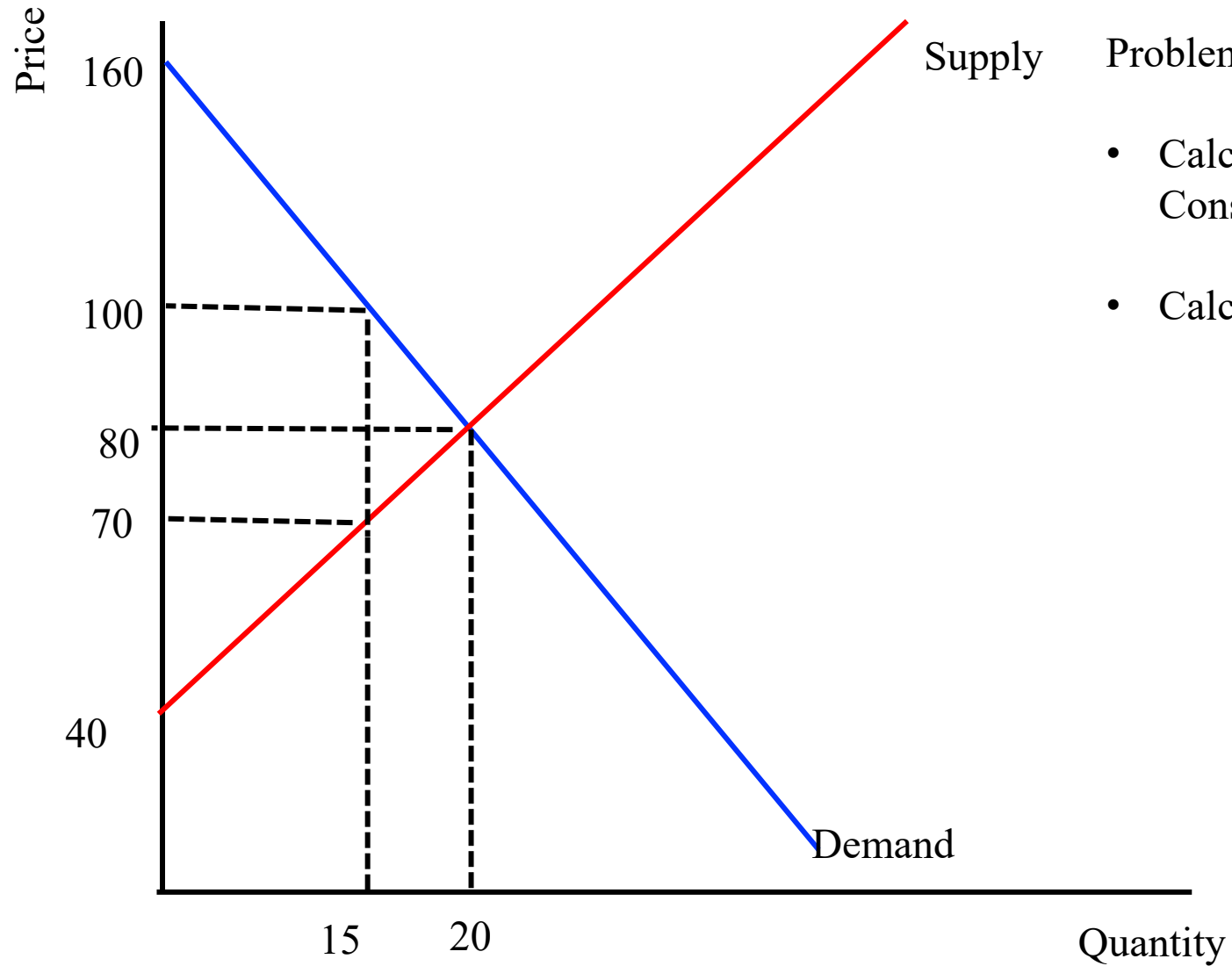
Practice Problems



Problem 2

- Calculate Producer Surplus and Consumer Surplus

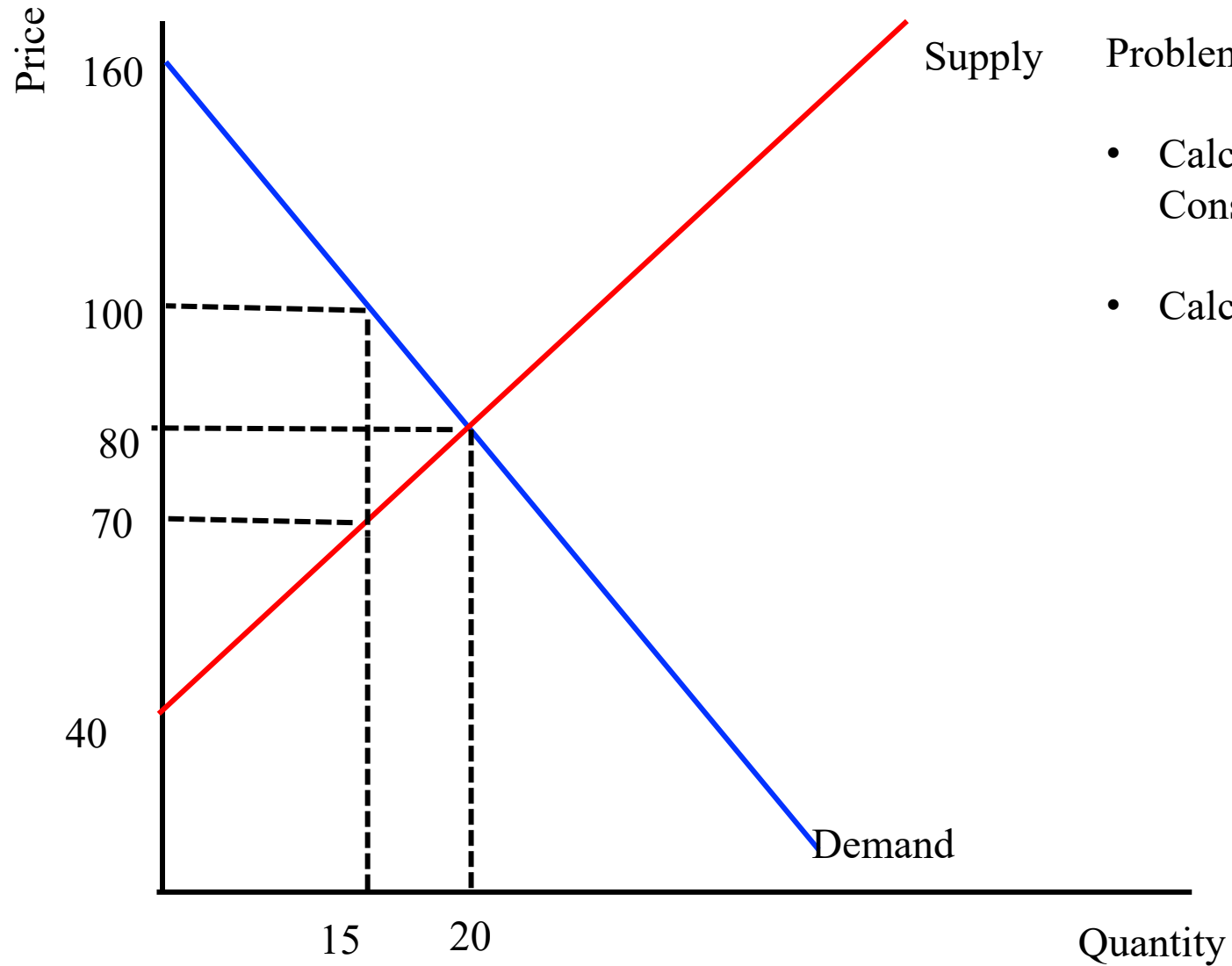
Practice Problems



Problem 2

- Calculate Producer Surplus and Consumer Surplus
- Calculate Tax Revenue

Practice Problems



Supply

Problem 2

- Calculate Producer Surplus and Consumer Surplus
- Calculate Tax Revenue

Demand

Credits to @grebcomics and Cowen and Tabarrok (2014)